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BCSE415L

Human Computer Interaction

DIGITAL ASSIGNMENT-2

**Personalized Gesture Vocabulary Builder: A Few-Shot
Learning System for Augmentative and Alternative
Communication**

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1. Introduction

Digital Assignment 1 finalized the topic and objectives for the Personalized Gesture Vocabulary Builder (PGVB), a few-shot, training-free AAC tool that lets a nonverbal or minimally verbal user, or a caregiver acting for them, teach a webcam-based system a personal hand gesture from a few demonstrations. This assignment reports the evaluation of the built prototype against the six objectives (O1 to O6) set out in DA-1, with every number below produced by a script from a committed recording, dataset, or config file in the project repository, not estimated by hand.

Evaluation covers three layers: the frozen embedding backbone trained on the Kaggle ASL Alphabet dataset (a proxy task used only to build a general hand-pose embedding space, never the personal gestures themselves), the open-set recognition threshold calibrated against real recordings, and a live participant evaluation (one team member, four personal gestures) covering recognition accuracy, false triggers, latency, stability across enrollment, and enrollment usability.

2. Backbone Few-Shot Accuracy and Distance Gap

The embedding network was trained once with a prototypical-network objective on ASL Alphabet landmarks and then frozen. Five letters (B, C, L, V, Y) were held out of training entirely and used here as never-seen poses, with support examples from one signer and query examples from a different signer, over 500 episodes.

Metric	Value	Target	Met
5-way 5-shot accuracy	0.988	≥ 0.85 (target 0.90)	Yes
5-way 1-shot accuracy	0.957	≥ 0.70	Yes
Mean intra-pose cosine distance	0.0960	Lower than inter-pose	Yes
Mean inter-pose cosine distance	1.0405	Higher than intra-pose	Yes
95th percentile intra-pose distance	0.9570	Informational	--

Source: `scripts/eval_backbone.py` -> `reports/backbone_v1_eval.md`.

3. Threshold Calibration Summary

A 200-trial sweep on the held-out ASL letters alone picked `max_distance = 0.39`, but this did not transfer to real personal gestures: replaying an idle-hands recording at 0.39 produced 22 false triggers, because the ASL-letter sweep never saw an idle-hand class to calibrate against. The threshold was re-swept directly against the team's own recordings and set to 0.10 in `configs/default.yaml`, the value used for every recognition result in this document.

Sweep	<code>max_distance</code>	Enrolled accuracy	False accept rate
ASL held-out letters (cross-signer)	0.39	0.983	0.044

Sweep	max_distance	Enrolled accuracy	False accept rate
Team recordings (used in configs/default.yaml)	0.10	See Section 5	See Section 5

Source: `scripts/calibrate_threshold.py` -> `reports/threshold_sweep.md`.

4. Runtime Performance

Stage	Value	Target	Met
Camera fps, 640x480, hand in frame	14.6 to 30.0 (tracker-bound)	>= 20 fps	Partial
Landmark tracking (track_ms)	27 to 31 ms live	Dominates per-frame time	--
Embed time (backbone_v1.npz)	0.033 ms mean, 1000 calls	< 1 ms	Yes
Recognize + smooth + trigger, excluding tracking	Well under 1 ms mean	< 25 ms	Yes

Source: `docs/phases/phase-1.md`, `docs/phases/phase-2.md`, `docs/phases/phase-3.md`; per-stage timing printed by `scripts/demo_recognize.py` at exit.

5. Recognition Accuracy, False Triggers, and Latency

One participant (kanadb) enrolled four personal gestures through the desktop app, 8 examples each: open_palm -> "Hello", peace_sign -> "Thank you", fist -> "I need help", thumbs_up -> "Okay". Three five-second clips per gesture plus one 60-second non-gesture clip (resting, typing, scratching, drinking) were recorded and replayed through the full pipeline.

Participant	Gestures	Clip accuracy	False triggers/min	Algorithmic latency (median)
kanadb	4	0.750 (9/12)	0.000	729.6 ms

Gesture	Clip accuracy
open_palm	1.000 (3/3)
peace_sign	1.000 (3/3)
fist	1.000 (3/3)
thumbs_up	0.000 (0/3)

Targets: clip accuracy ≥ 0.85 (not met, 0.750, entirely due to thumbs_up), false triggers/min < 1 (met, 0.000), algorithmic latency median < 600 ms (not met, 729.6 ms).

- thumbs_up is a false reject, not a misrecognition: every thumbs_up frame's nearest prototype was correctly thumbs_up, but the mean cosine distance to its own prototype on the eval clips (0.134) exceeded recognize.max_distance (0.10). thumbs_up and fist were both markedly slower and less stable to enroll than the two open-hand gestures (Section 7), producing a looser prototype fit.
- Algorithmic latency exceeded the 600 ms target because these clips ran at roughly 15 to 16 fps: smoothing.min_votes = 8 alone needs about 8/15.5 s (~515 ms) of continuously correct decisions to confirm a label at this frame rate, before any pose ramp-up time, consistent with the measured 729.6 ms median.

Trigger-to-speech latency (live, demo_recognize.py --speak): median 0.1 ms over 50 triggers. This measures the Speaker worker thread's queue-to-dequeue latency, which fires before the TTS backend is invoked, not the audible onset of speech; it is expected to be near-instantaneous and is not itself a measure of perceived responsiveness.

Source: scripts/eval_recognition.py -> reports/eval_recognition_kanadb.md/.csv; figures in reports/img/kanadb_confusion.png and reports/img/kanadb_latency.png.

6. Stability Across Enrollment (O4)

Gestures were added back into the profile one at a time in their original enrollment order (open_palm, peace_sign, fist, thumbs_up), and after each addition every already-enrolled gesture was re-evaluated on its own clips.

Gestures enrolled	open_palm	peace_sign	fist	thumbs_up
1	1.000	--	--	--
2	1.000	1.000	--	--
3	1.000	1.000	1.000	--
4	1.000	1.000	1.000	0.000

O4 assertion: passed. Every gesture's prototype was bit-identical to the fully enrolled profile's prototype at every earlier enrollment step; thumbs_up's 0.000 accuracy is the same threshold issue as Section 5, not a stability failure, and enrolling it did not change the other three gestures' accuracy.

Source: scripts/eval_stability.py -> reports/eval_stability_kanadb.md; figure reports/img/kanadb_stability.png.

7. Usability: Enrollment Time and Observations

Gesture	Enrollment time
open_palm	5 s

Gesture	Enrollment time
peace_sign	10 s
fist	~105 s (1.5 to 2 min, timed as a range)
thumbs_up	~105 s (1.5 to 2 min, timed as a range)

- The two open, flat-hand gestures enrolled in 5 to 10 seconds: the pose settles (passes the stability-gate motion check) almost immediately once held.
- The two closed-hand gestures took 10 to 20 times longer to enroll: small, involuntary finger movement in a closed fist, or a thumb held against one, keeps the landmark motion above the enrollment stability threshold longer than an open palm does. This is a caregiver-usability problem, not a training-free-recognition problem.
- The same instability at enrollment time is very likely why thumbs_up's later eval clips sit further from its own prototype than the threshold tolerates (Section 5): a slower, less consistent enrollment produced a less representative prototype. Fixing enrollment stability detection for closed-hand poses, or requiring more samples when intra-sample distance is high during enrollment, would likely address both findings together; noted as backlog rather than fixed in this cycle.

Source: `data/sessions/eval/kanadb/enrollment_times.csv`.

8. Results Against Objectives (O1 to O6)

Objective	Result
O1: real-time landmark extraction	Met
O2: few-shot enrollment, no retraining	Met
O3: open-set rejection	Met
O4: stability (enrolling gesture N never disturbs 1..N-1)	Met
O5: spoken and on-screen output	Met
O6: real-time responsiveness and enrollment usability	Partially met

O1, O2, O3, O4, and O5 are fully met, verified both by unit tests on the pure math (features, prototypes, smoothing, embedding parity, stability) and by running the evaluation scripts against committed recordings. O6 is partially met: recognition is real-time in the sense of continuous per-frame processing, but algorithmic latency (729.6 ms) exceeds the 600 ms target at this webcam's tracker-bound frame rate, and enrollment usability was uneven across gesture shapes, fast for open-hand poses and slow for

closed-hand poses. Both findings are explained above rather than hidden, and both point to the same root cause: the enrollment stability gate is not yet well tuned for closed-hand poses.

9. Working Demo Video – Google Drive Link

<https://drive.google.com/file/d/1wjzgFdCo1lJvxKZfjloc3d0DxA8KwW5j/view?usp=sharing>

10. Conclusion

The prototype meets its core technical objectives: it extracts hand landmarks in real time, enrolls a new personal gesture from a handful of examples with no retraining, rejects unenrolled poses, never disturbs an earlier gesture's recognition when a new one is added, and responds with speech and an on-screen message. The one participant evaluation surfaced a genuine, reproducible limitation, closed-hand poses are slower to enroll and can end up false-rejected at the calibrated threshold, which is recorded honestly here and carried into the project Backlog rather than concealed by re-recording until a better-behaved sample was produced.